DUAL-AXIS 8-LAYER AUTONOMOUS AGENTIC VERIFICATION & GOVERNANCE ARCHITECTURE (HQAS-8)

Inventor & Principal Systems Architect: HM Shihabuddin Jashory (Shihab) | ORCID: 0009-0003-4163-0092 | Vision Flow Research Lab (2026)

HM SHIHABUDDIN JASHORY (SHIHAB) -- SUPREME SYSTEMS COMMANDER

Principal Scientist & AI Systems Architect | ORCID: 0009-0003-4163-0092

(Issues High-Level Executive Directive & Receives 100% Deterministically Verified Final Output)

INTER-TIER 4: SUPREME INSPECTOR GENERAL AGENT

Single Executive Command Hub & Cross-Pipeline Verification Gatekeeper

(Routes Tasks to Parallel Project Pipelines & Blocks Unverified Outputs)

PARALLEL PROJECT PIPELINE A (e.g., Video / Multimedia / UI Engine)

INTER-TIER 3: CHIEF AUDITOR AGENT A

Audits BOTH Supervisor & Worker transcript.jsonl + Verifies Disk Frames/Audio vs. Primary Source Spec

INTER-TIER 2: SUPERVISOR AGENT A (45s-60s Heartbeat)

1. Tail-Reads Worker transcript.jsonl every 45s-60s | 2. Visual/Code Disk Audit | 3. In-Flight Correction (send_message)

INTER-TIER 1: WORKER AGENT A (Heavy Execution)

Executes Browser / Code / FFmpeg / Raw Clips / Stems / MP4 & CapCut Draft in Isolated Context

PARALLEL PROJECT PIPELINE B (e.g., Offensive Security / NASA VDP / Cloud)

INTER-TIER 3: CHIEF AUDITOR AGENT B

Audits BOTH Supervisor & Worker transcript.jsonl + Verifies PoC / Advisories vs. Primary Source Spec

INTER-TIER 2: SUPERVISOR AGENT B (45s-60s Heartbeat)

1. Tail-Reads Worker transcript.jsonl every 45s-60s | 2. PoC & Build Disk Audit | 3. Phase-by-Phase Atomic Gate

INTER-TIER 1: WORKER AGENT B (Heavy Execution)

Executes Vulnerability Scanning / Code / Cloud VM / Certification Automation in Isolated Context

AXIS A -- 4 INTRA-AGENT DETERMINISTIC GOVERNANCE GUARDS (ENFORCED INSIDE EVERY SINGLE AGENT TIER)

1. Zero-Repeat Bootstrap

START_HERE.md & CONTEXT.md
Auto-loads full project state & architecture without human repetition.

2. Domain Rules & Skills

config/rules & config/skills
Enforces Video QC, Security Isolation, Email Gate & Identity Firewall.

3. Self-Evolving Memory

hermes_memory_engine.py
Permanent anti-regression ledger (hermes_ledger.json) across chats.

4. Python Hard Block Hooks

shihab_enforcer.py & video_qc_gate.py
PreToolUse & Stop Hooks block unverified outputs (decision='block').

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